Chemical Kinetics

(10/16/2015)

If A reacts to form either B or C according to

$$A \xrightarrow{k_1} B \quad \text{or} \quad A \xrightarrow{k_2} C$$

(1) show that $[A] = [A]_0 e^{-(k_1+k_2)t}$

(2) show that $t_{1/2}$, the half-life of A, is given by $t_{1/2} = \dfrac{0.693}{k_1 + k_2}$

(3) show that [B]/[C] = k_1 / k_2 for all times t

(4) show that E_a, the observed activation energy for the disappearance of A, is given by $E_a = \dfrac{k_1 E_1 + k_2 E_2}{k_1 + k_2}$, where E_1 is the activation energy for the first reaction and E_2 is the activation energy for the second reaction. (Hint: The temperature dependence of the rate constants can be described by the Arrhenius equation $k = Ae^{-E_a/RT}$.)

(5) Potassium-argon dating is used in geology and archeology to date sedimentary rocks. Potassium-40 decays by two different paths analog to those reactions shown above (A reacts to form either B or C):

$${}^{40}_{19}K \rightarrow {}^{40}_{20}Ca + {}^{0}_{-1}e \qquad \text{(89.3\% by mol)}$$

$${}^{40}_{19}K \rightarrow {}^{40}_{18}Ar + {}^{0}_{1}e \qquad \text{(10.7\% by mol)}$$

The overall half-life for the delay of potassium-40 is 1.3×10^9 y. Estimate the age of sedimentary rocks with an argon-40 to potassium-40 molar ratio of 0.0102.